Venkata Lakshmi Parimala Pasupuleti

Agentic Analytics Engineer · Python · SQL · AI Agents · LLMs · Analytics Engineering · Data Engineering · PropTech · Real Estate Domain · Production AI

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA — open to Seattle / Miami

PROFESSIONAL SUMMARY

Agentic analytics engineer with 8 years building data systems and production AI at an Opendoor competitor (Entrata) — the same real estate domain, same property managers, same financial workflows. Built **AgentProbe**: autonomous AI agents (LangGraph, Claude API) that surface outcomes without manual initiation — proactive intelligence with eval gates, observability, and CI/CD regression tracking. Built **Leasel**: agentic RAG pipeline with a **semantic layer for LLM reasoning** over 1,240 legal documents (93.6% precision), natural language compliance queries, and a deterministic rules engine for fully auditable outputs. At Entrata: **expert-level SQL** across 200+ production query optimizations (30% performance gain), data-driven automation eliminating 80% of manual workflows, financial data pipelines on 1.6M records. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — Analytics Engineering · ML Pipelines · Data Modeling · Stakeholder Intelligence

- Built multi-source data pipelines (Python, R, 190+ countries, WHO + UK Road Safety datasets) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines surfacing anomalies manual review missed; designed analytics dashboards (Tableau, ggplot2, Quarto) driving **40% improvement** in business decision comprehension for faculty leadership.

Entrata — Opendoor Competitor (PropTech / Real Estate)

Jan 2021 – Aug 2024

Senior Software Engineer — Data Engineering · Financial Analytics · Production AI · AWS · Mentorship

- Built and owned **financial data pipelines and analytics systems** for property managers and owners — owner financial reporting, distribution analytics, leasing data workflows, income eligibility data models — across 20+ production modules (**1M+ records, 100K+ MAU**, 99.5% uptime, 120+ Agile sprints).
- Delivered Entrata's **first production AI pipeline on 1.6M real financial records** (Python, PyTorch, Claude/OpenAI API) with output validation, quality gates, and proactive anomaly alerting before customer exposure; optimized **200+ SQL queries** (30% performance gain) across all client environments — execution plan tuning, index strategy, join optimization.
- Eliminated **80% of manual operational workload** by designing data-driven automation workflows (Python, PL/SQL); achieved **100% PHI compliance** on AWS S3/RDS/Glue data migration; mentored 5+ engineers, **300+ code reviews**.

Entrata

Jan 2017 – Dec 2020

Software Engineer / Associate Engineer — Full-Stack · Data Pipelines · SQL · API Integrations

- Built data pipelines and analytics layers processing **millions of records** — income thresholds, compliance data models, financial account workflows; 200+ peer code reviews; 0→1 feature delivery through production rollout across 100K+ unit portfolios.

PROJECTS

AgentProbe: Autonomous Multi-Agent Analytics & Eval Framework

AI AGENTS · PROACTIVE INTELLIGENCE

Python · LangGraph · Claude API · OpenAI · FastAPI · ChromaDB · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- **Autonomous agent runtime**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) surfaces outcomes without manual initiation — proactive intelligence with tool calling, inter-agent state handoffs, and multi-provider LLM routing (Claude, OpenAI, Groq, Ollama); CI/CD regression gates fire automatically on every commit; React **observability dashboard** tracks per-run metrics, latency, and retrieval drift without human query.

Leasel: Agentic RAG with Semantic Layer for Legal Document Reasoning

PROPTECH · SEMANTIC LAYER

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 residential lease PDFs — built a **semantic layer optimized for LLM reasoning** over real estate documents: extraction, chunking, embedding, retrieval, natural language compliance queries; deterministic rules engine for fully auditable outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%).

CrashClarity: Large-Scale Analytics on 104K Road Safety Records

ANALYTICS ENGINEERING

Python · R · scikit-learn · Random Forest · Feature Engineering · Tableau · GitHub Pages — Portfolio

- **85%+ accuracy** on 104K+ UK accident records — end-to-end analytics engineering: data modeling, feature engineering (weather, road, time variables), Random Forest + Logistic Regression pipeline, interactive Tableau-style dashboard on GitHub Pages driving **actionable insights** from raw data.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Analytics engineering, ML systems, statistical modeling, data pipelines, AI systems, data governance, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Analytics: Python, SQL (expert — execution plan tuning, index strategy, complex data modeling), PL/SQL, R, JavaScript, TypeScript; dbt (familiar), Airflow (familiar/adjacent - event-driven pipeline background), Snowflake (familiar), BigQuery (familiar), PostgreSQL, MySQL

AI Agents & LLMOps: LangGraph, LangChain, Agentic AI, Autonomous Agent Workflows, Multi-Agent Orchestration, Proactive Intelligence Systems, Natural Language Interfaces, Semantic Layers, RAG Pipelines, Tool Calling, LLM-as-Judge, LLM Evaluation, Prompt Engineering, Claude API (Anthropic), OpenAI API, ChromaDB, pgvector, PyTorch, BERT, NLP

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Microservices, data warehouse, ETL/ELT, Tableau, Power BI, ggplot2, HIPAA/PHI compliance, Agile/Scrum, Git, Linux